

Duopoly

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Duopoly: Market in which two firms compete with each other.

The Cournot Model: An oligopoly model first introduced by the French economist Augustin Cournot in 1838 in which firms produce a homogeneous good, each firm treats the output of its competitors as fixed, and all firms decide simultaneously how much to produce.

Assumption of Cournot model :-

- (i) Every firm will produce homogeneous products.
- (ii) Each firm knows the market demand.
- (iii) Each firm must decide how to produce and the two firms make their decision at the same time.
- (iv) Market price depends on total output.
- (v) When a firm makes a production decision, the firm will consider the decision taken by other.

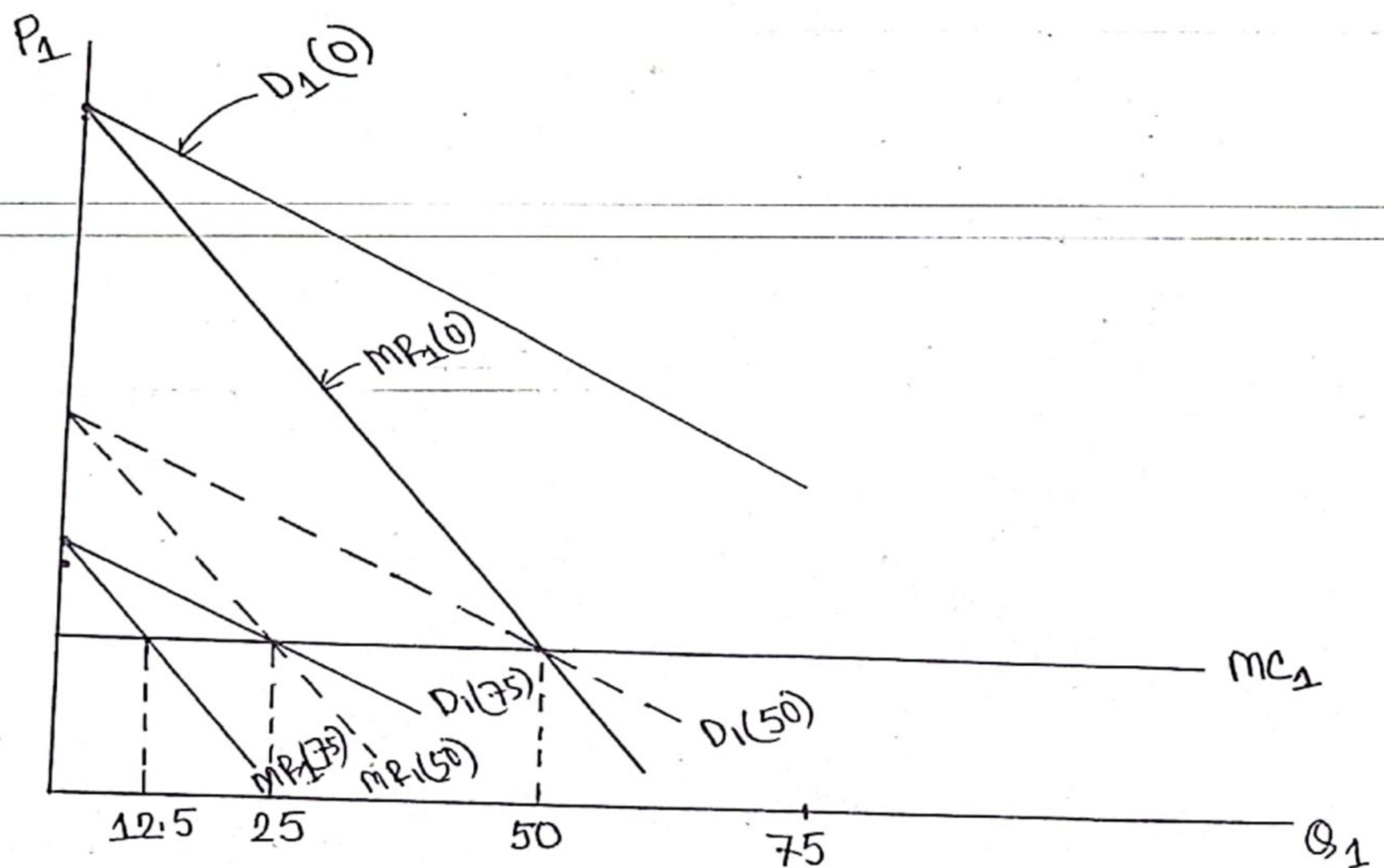
- (vi) Each firm knows that its competitor is also deciding how much to produce.
- (vii) Marginal cost is constant.

Suppose, there are two firms

(i) Firm-1 (F_1)

(ii) Firm-2 (F_2)

now, suppose F_1 thinks that F_2 will produce nothing. In that case F_1 's demand curve is the market demand curve.



In this figure $D_1(0)$ is the demand curve for F_1 , assuming F_2 produces zero. $MR_1(0)$ is the MR curve for F_1 , we have assumed that MC_1 is constant.

F_1 's profit maximizing output is 50 units because at this output $MC_1 = MR_1(0)$.

If F_1 thinks F_2 will produce 50 units, then F_1 's demand curve is the market demand curve shifted to the left by 50. Here F_1 's demand curve is $D_1(50)$ and MR curve is $MR_1(50)$. So F_1 's profit maximizing output is 25 units where $MR_1(50) = MC_1$.

Again if F_1 thinks F_2 will produce 75 units, then its demand curve will be $D_1(75)$ and MR curve is $MR_1(75)$. So the profit maximizing output is 12.5 unit where $MR_1(75) = MC_1$.

If firm-1 thinks F_2 will produce 100 units then F_1 will produce nothing.

Now, we can draw the reaction curves of both firms.

reaction curve: Relationship between a firm's profit maximizing output and the amount it thinks its competitor will produce.

If F_1 thinks that F_2 will produce nothing, it will produce 50; if it thinks F_2 will produce 50, it will produce 25; if it thinks F_2 will produce 75, it will produce 12.5; and if it thinks F_2 will produce 100, then it will produce nothing. F_1 's profit-maximizing output is thus a decreasing schedule of how much it thinks F_2 will produce. We call this schedule F_1 's reaction curve and denote it by $Q_1^*(Q_2)$.

With the same analysis we have the F_2 's reaction curve $Q_2^*(Q_1)$.

The equilibrium output levels is therefore found by the intersection of the two firms' reaction curves.

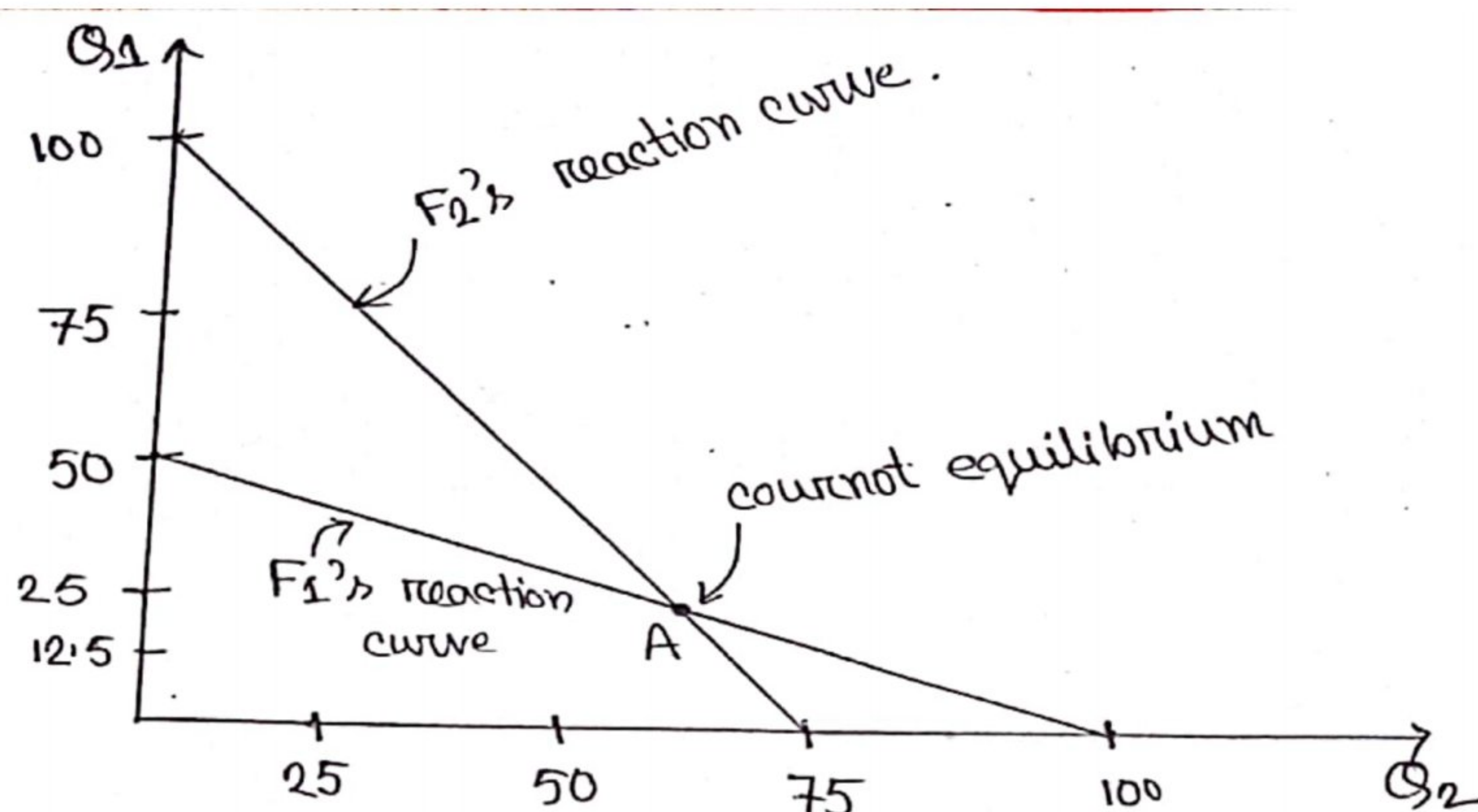


Figure: Reactions curves and Cournot equilibrium

Here, Point A is the equilibrium which is known as Cournot equilibrium. In this equilibrium, each firm correctly assumes how much its competitor will produce and sets its own production level accordingly.

This Cournot equilibrium is an example of Nash equilibrium so that it is sometimes called a Cournot-Nash equilibrium.

*** Find the Cournot equilibrium and compare it with the perfect competitive equilibrium and output when firms collude, from the following information:

$$P = 30 - Q$$

$$Q = Q_1 + Q_2$$

$$MC_1 = MC_2 = 0$$

We know that,

$$TR_1 = PQ_1 = (30 - Q_1 - Q_2)Q_1 = 30Q_1 - Q_1^2 - Q_1Q_2$$

$$\therefore MR_1 = 30 - 2Q_1 - Q_2$$

At the profit maximizing level, $MR_1 = MC_1$

$$30 - 2Q_1 - Q_2 = 0$$

$$\Rightarrow 2Q_1 = 30 - Q_2$$

$$\Rightarrow Q_1 = 15 - \frac{1}{2}Q_2 \quad \dots \quad \textcircled{1}$$

This is firm-1's reaction curve.

Again, $TR_2 = PQ_2$

$$= (30 - Q_1 - Q_2)Q_2 = 30Q_2 - Q_1Q_2 - Q_2^2$$

$$\therefore MR_2 = 30 - Q_1 - 2Q_2$$

At the profit maximizing level, $MR_2 = MC_2$

$$\therefore 30 - Q_1 - 2Q_2 = 0$$

$$\Rightarrow 2Q_2 = 30 - Q_1$$

$$\therefore Q_2 = 15 - \frac{1}{2}Q_1 \quad \dots \quad (2)$$

This is firm-2's reaction curve.

now, from equation (1) we can write,

$$Q_1 = 15 - \frac{1}{2} \left(15 - \frac{1}{2}Q_1 \right)$$

$$= 15 - \frac{15}{2} + \frac{1}{4}Q_1$$

$$\Rightarrow Q_1 - \frac{1}{4}Q_1 = \frac{15}{2}$$

$$\Rightarrow \frac{3Q_1}{4} = \frac{15}{2}$$

$$\Rightarrow 3Q_1 = 30$$

$$\therefore Q_1 = 10$$

$$\therefore Q_2 = 15 - \frac{1}{2} \times 10 = 10$$

Thus, the equilibrium output is $(Q_1, Q_2) = (10, 10)$ which is determined by the intersection of both firm's reaction curve.

$$\therefore \text{Cournot equilibrium: } Q_1 = Q_2 = 10.$$

So, total output $Q = 10 + 10 = 20$.

Thus, the equilibrium market price is,

$$P = 30 - Q = 30 - 20 = 10$$

So, Profit, $\pi = PQ - TC$

$$= (10 \times 20) - 0$$

$$= 200$$

Thus, each firm's profit is, $\pi_1 = \pi_2 = 100$.

In perfect competition;

$$P = MC$$

$$\Rightarrow 30 - Q = 0$$

$$\Rightarrow Q = 30$$

So, each firm will produce $Q_1 = Q_2 = 15$ units.

So, each firm earns a zero-economic profits.

If firm's collude then,

$$MC = MR$$

$$TR = PQ$$

$$= (30 - Q)Q = 30Q - Q^2$$

...

$$MR = 30 - 2Q$$

$$\therefore MC = MR$$

$$\text{thus, } 30 - 2Q = 0$$

$$\Rightarrow Q = 15$$

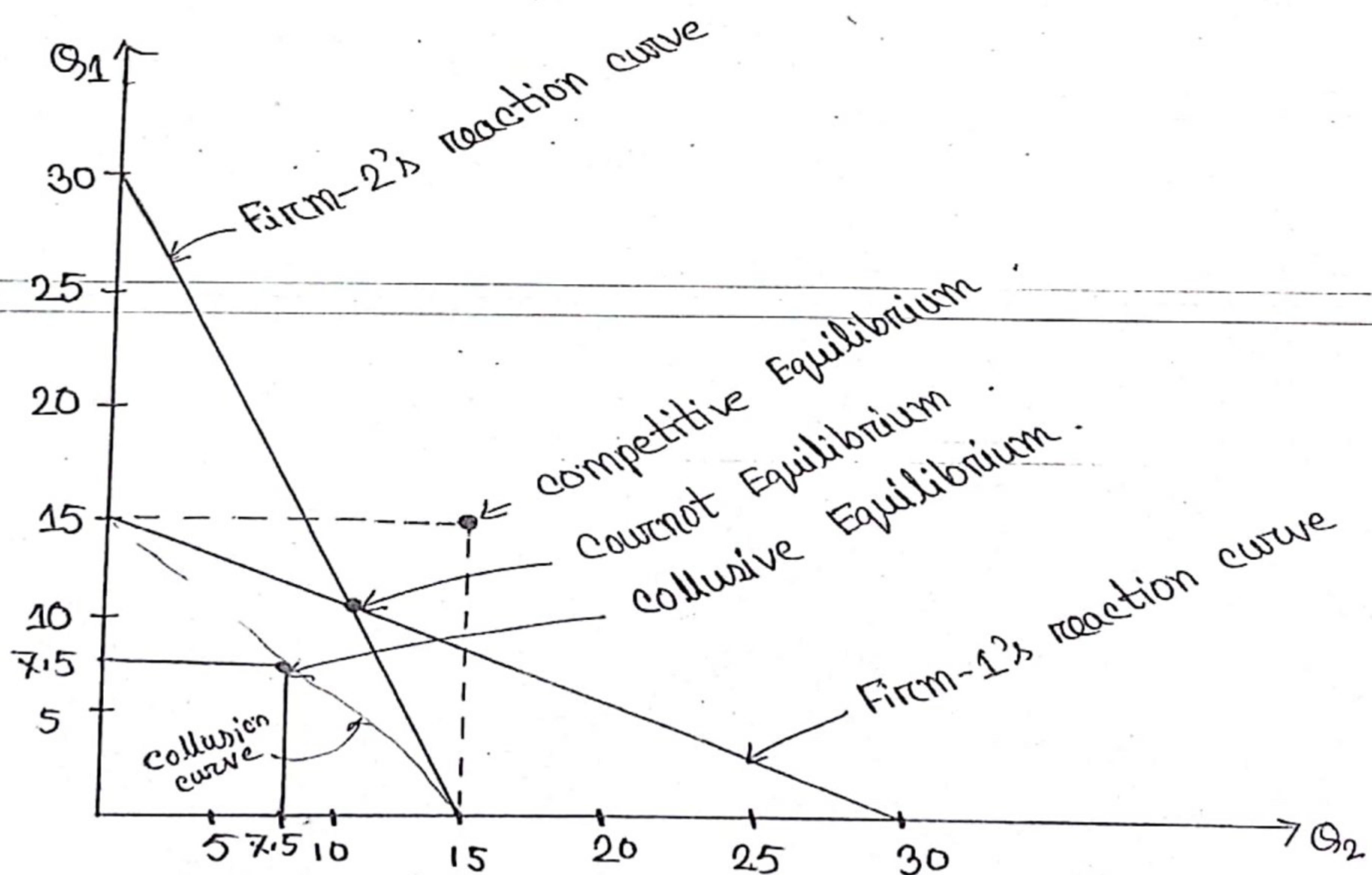
$$\therefore Q_1 = Q_2 = 7.5$$

$$\text{Thus, profit } \pi = TR - TC$$

$$= 30Q - Q^2 - 0$$

$$= (30 \times 15) - (15)^2 = 225$$

$$\therefore \pi_1 = \pi_2 = 112.5$$



By comparing these three equilibrium we can say that ,

Cournot outcome is much better than perfect competition but not as good as the outcome from collusion .

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Stackelberg model: First Mover Advantage: Oligopoly
model in which one firm sets its output before other firms do.

Assumptions: Let there are two firms, F-1 and F-2.
(i) Firms do not take the decision at the same time, one firm set its output first.

(ii) After observing firm-1's output decision, F-2 will take the decision.

(iii) F-1 will consider how F-2 will react.

(iv) Neither of the firms have any opportunity to change its reaction.

Let, F-1 will make the decision first and then F-2 will decide.

Given, $P = 30 - Q$ and $Q_1 + Q_2 = Q$

$$MC_1 = MC_2 = 0$$

From the Cournot model we can write,

$$Q_2 = 15 - \frac{1}{2} Q_1 \quad \text{--- (1)}$$

which is firm-2's reaction curve.

Firm-1 will set output at Q_1 , in which he can

maximize his profit.

$$\therefore TR_1 = PQ_1 = (30 - Q_1 - Q_2)Q_1 = 30Q_1 - Q_1^2 - Q_1Q_2 \quad \text{--- (1)}$$

As R_1 depends on Q_2 , Firm-1 must anticipate how much Firm-2 will produce. F_1 has a clear idea about F_2 's reaction curve. Now substituting equation (1) for Q_2 into equation (1), we find that F_1 's revenue is

$$R_1 = 30Q_1 - Q_1^2 - Q_1(15 - \frac{1}{2}Q_1)$$

$$= 30Q_1 - Q_1^2 - 15Q_1 + \frac{1}{2}Q_1^2$$

$$= 15Q_1 - \frac{1}{2}Q_1^2$$

$$\therefore MR_1 = 15 - Q_1$$

At profit maximizing level, $MR_1 = MC_1 = 0$

$$15 - Q_1 = 0$$

$$\therefore Q_1 = 15$$

$$\text{Thus, } Q_2 = 15 - \frac{1}{2} \times 15 = 7.5$$

$$\therefore \text{Price } P = 30 - (Q_1 + Q_2) = 30 - (15 + 7.5) = 7.5$$

$$\text{therefore, } \pi_1 = PQ_1 - TC = (7.5 \times 15) - 0 = 112.5$$

$$\pi_2 = PQ_2 - TC = (7.5 \times 7.5) - 0 = 56.25$$

So, F_1 produces twice as much as Firm 2 and makes twice as much profit. Going first gives Firm 1 an advantage.

Bertrand model: Bertrand model is an oligopoly model in which firms produce a homogeneous good, each firm treats the price of its competitors as fixed, and all firms decide simultaneously what price to charge. The Bertrand model was developed by another French economist, Joseph Bertrand in 1883.

Price Competition with Homogeneous Products - The Bertrand Model;

Assumptions:

- (i) Each firm produces homogeneous products.
- (ii) Each firm takes the decision at the same time.
- (iii) They decide what price to charge instead of quantity.

Suppose, there are two firms: F-1 and F-2.

Let, market demand is,

$$P = 30 - Q, \text{ where } Q = Q_1 + Q_2$$

$$MC_1 = MC_2 = \$3$$

As the good is homogeneous, consumer will purchase

only from the lowest price seller, if two firms charge different price, the lower-price firm will supply the entire market and the higher price firm will sell nothing. So, both firm will charge same price and consumer will be indifferent as to which firm they buy. So, each firm will supply half the market.

Both firm set the price at $P = MC$.

Thus $P_1 = P_2 = MC_1 = MC_2 = \3

Now, $P = 30 - Q$

$$\Rightarrow 30 - Q = 3 \Rightarrow Q = 27$$

$$\therefore \pi_1 = P_1 Q_1 - MC_1(Q_1) = (3 \times 13.5) - (3 \times 13.5) \\ = \text{Zero economic profit}$$

$$\pi_2 = P_2 Q_2 - MC_2(Q_2) = (3 \times 13.5) - (3 \times 13.5) \\ = \text{Zero economic profit}$$

There couldn't be a nash equilibrium in which the firm charged the same price, but a higher one,

So that each made some profit, because if either one firm lowered its price just a little, it could capture the entire market and nearly double its profit. Thus, each firm would want to undercut its competitor. Such undercutting would continue until the price dropped to \$ 3.

Criticism: The Bertrand model has been criticized on several counts.

First: when firms produce a homogeneous good, it is more natural to compete by setting quantities rather than prices.

Second: Firms can produce different amount of output. There is no reason to assume that sales would be divided equally among the firms.

Price Competition with Differentiated Products;

Oligopolistic markets often have at least some degree of product differentiation. Market shares are determined not just by prices, but also by differences in the design, performance, and durability of each firm's product. In such cases, it is natural for firms to compete by choosing prices rather than quantities.

Let,

$$F-1's \text{ demand } Q_1 = 12 - 2P_1 + P_2$$

$$F-2's \text{ demand } Q_2 = 12 - 2P_2 + P_1$$

where P_1 and P_2 are the prices that Firms 1 and 2 charge, respectively, and Q_1 and Q_2 are the resulting quantities that they sell.

assumptions:

(i) Fixed cost = \$20, variable cost = 0

(ii) Both firms set their prices at the same time and consider competitor's price as fixed.

$$\begin{aligned} \text{Now, } \pi_1 &= P_1 Q_1 - (FC + VC) = P_1 Q_1 - 20 \\ &= 12P_1 - 2P_1^2 + P_1 P_2 - 20 \end{aligned}$$

Firm-1 assumes that, P_2 is fixed. So Firm-1's profit maximizing price is therefore given by

$$\frac{\partial \pi_1}{\partial P_1} = 12 - 4P_1 + P_2 = 0$$

$$\Rightarrow 4P_1 = 12 + P_2$$

$$\therefore P_1 = 3 + \frac{1}{4}P_2$$

This is the Firm-1's reaction curve.

Similarly we can find the firm-2's reaction curve which is :

$$P_2 = 3 + \frac{1}{4}P_1$$

$$\text{Now, } P_1 = 3 + \frac{1}{4}(3 + \frac{1}{4}P_1)$$

$$\Rightarrow P_1 = 3 + \frac{3}{4} + \frac{1}{16}P_1$$

$$\Rightarrow P_1 - \frac{1}{16}P_1 = \frac{15}{4}$$

$$\Rightarrow \frac{15P_1}{16} = \frac{15}{4}$$

$$\Rightarrow 15P_1 = 60$$

$$\therefore P_1 = 4$$

$$\text{and } P_2 = 3 + \frac{1}{4} \times 4 = 3 + 1 = 4$$

Thus, $\pi_1 = 12(4) - 2(4)^2 + 4 \times 4 - 20$
 $= 48 - 32 + 16 - 20$
 $= \$ 12$

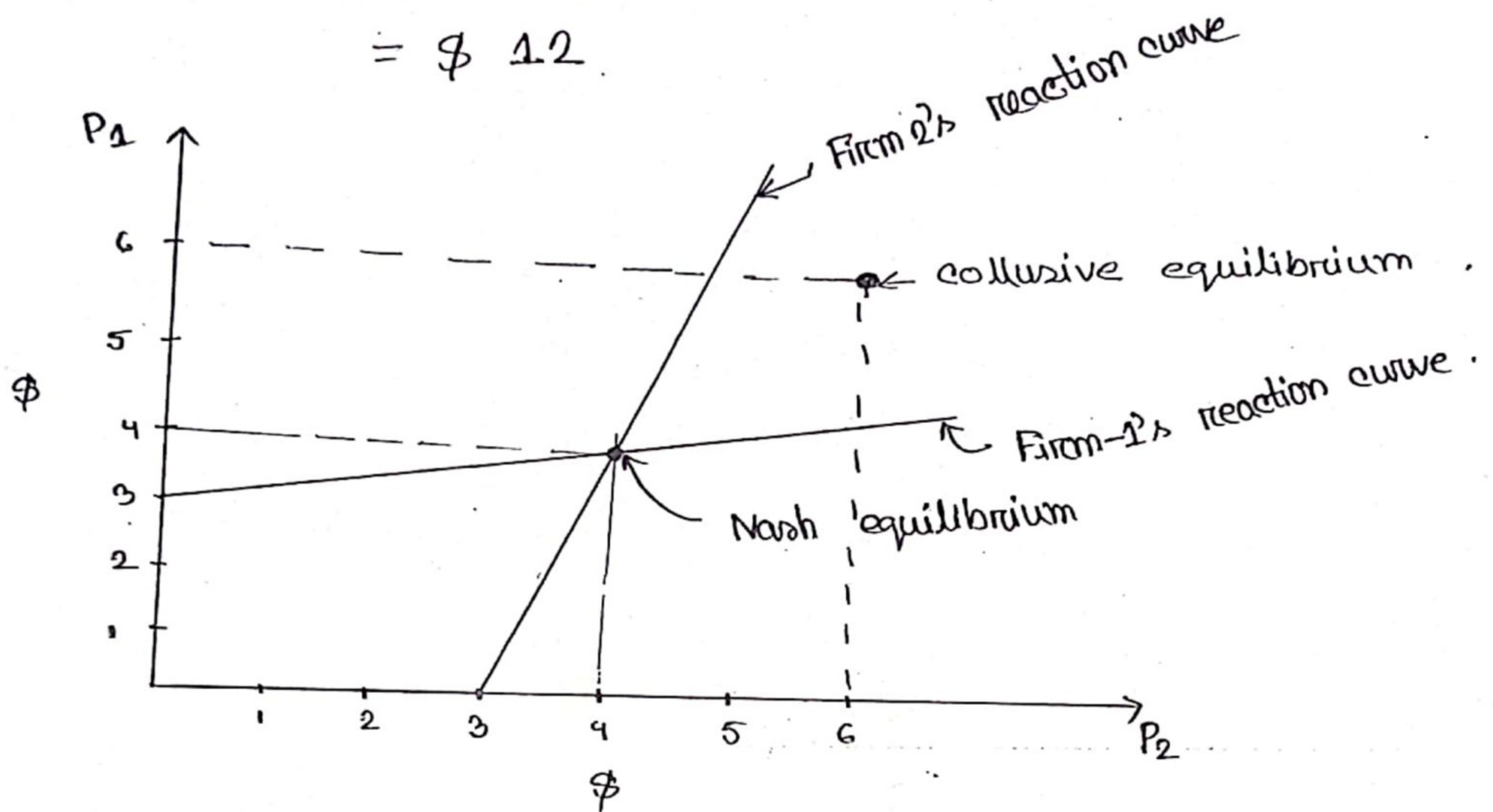


Figure: Nash equilibrium in Prices.

The reaction curves are drawn in figure. The Nash equilibrium is at the point where two reaction curve cross and it is the point where is firm then charging a price of \$4 and earning a profit of 12. At this point, because each firm is doing the best it can given the price its competitor has set, neither firm has an incentive to change its price.

Now suppose the two firms collude: Instead of choosing their prices independently, they both decide to charge the same price - namely the price that maximizes both of their profits.

Now, total Profits $\pi_T = \pi_1 + \pi_2$

$$\pi_1 = 12P - 2P^2 + P^2 - 20 = 12P - P^2 - 20 = \pi_2$$

$$\begin{aligned}\therefore \pi_T &= 12P - P^2 - 20 + 12P - P^2 - 20 \\ &= 24P - 2P^2 - 40\end{aligned}$$

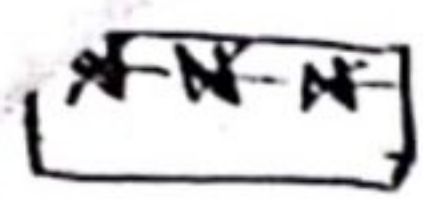
$$\therefore \frac{\partial \pi_T}{\partial P} = 24 - 4P = 0$$

$$\Rightarrow 4P = 24$$

$$\therefore P = 6$$

$$\therefore \pi_T = 24(6) - 2(6)^2 - 40 = 144 - 72 - 40 = 32$$

$$\therefore \pi_1 = \pi_2 = \frac{32}{2} = \$ 16$$



Find the Cournot equilibrium and compare it with the perfect competitive equilibrium and output and when firms collude, from the given information:

$$P = 30 - Q$$

$$Q = Q_1 + Q_2$$

$$MC_1 = MC_2 = \$3$$

we know that,

$$R_1 = P Q_1 = (30 - Q_1 - Q_2) Q_1 = 30Q_1 - Q_1^2 - Q_1 Q_2$$

$$\therefore MR_1 = 30 - 2Q_1 - Q_2$$

At the profit maximizing level $MR_1 = MC_1$

$$\therefore 30 - 2Q_1 - Q_2 = 3$$

$$\Rightarrow 2Q_1 = 27 - Q_2$$

$$\Rightarrow Q_1 = \frac{27}{2} - \frac{1}{2} Q_2 \quad \dots \quad ①$$

this is Firm-1's reaction curve.

Similarly, we can find the Firm-2's reaction curve

which is

$$Q_2 = \frac{27}{2} - \frac{1}{2} Q_1 \quad \dots \quad ②$$

Substituting the value of Q_2 from equation ② into equation ① we get:

$$Q_1 = \frac{27}{2} - \frac{1}{2} \left(\frac{27}{2} - \frac{1}{2} Q_1 \right)$$

$$\Rightarrow Q_1 = \frac{27}{2} - \frac{27}{4} + \frac{1}{4} Q_1$$

$$\Rightarrow Q_1 - \frac{1}{4} Q_1 = \frac{27}{4}$$

$$\Rightarrow \frac{3Q_1}{4} = \frac{27}{4}$$

$$\Rightarrow 3Q_1 = 27$$

$$\therefore Q_1 = 9$$

$$\text{So, } Q_2 = \frac{27}{2} - \frac{1}{2} \times 9 = \frac{27-9}{2} = \frac{18}{2} = 9$$

$\therefore$ Cournot eqⁿ $\rightarrow Q_1 = Q_2 = 9$

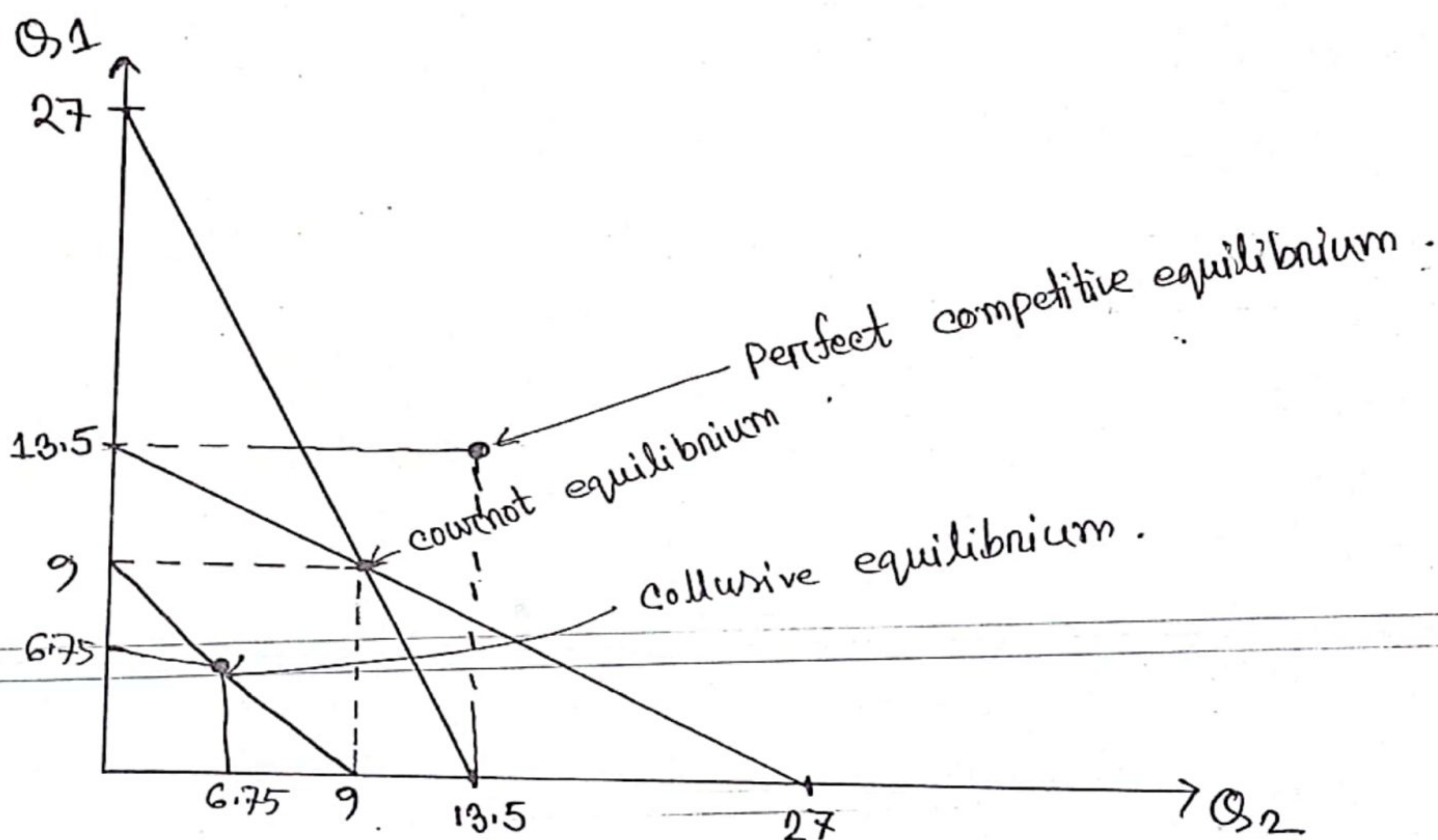
$$\text{Total output } Q = 9 + 9 = 18$$

Thus the equilibrium market price is

$$P = 30 - Q = 30 - 18 = \$12$$

$$\begin{aligned}
 \text{So Firm-1's profit } \pi_1 &= PQ_1 - MC_1(Q_1) \\
 &= (12 \times 9) - (3 \times 9) \\
 &= 108 - 27 \\
 &= 81
 \end{aligned}$$

$$\begin{aligned}
 \text{and Firm-2's profit, } \pi_2 &= PQ_2 - MC_2(Q_2) \\
 &= (12 \times 9) - (3 \times 9) \\
 &= 108 - 27 = 81
 \end{aligned}$$



In perfect competition :

$$P = MC$$

$$\Rightarrow 30 - Q = 3$$

$$\Rightarrow Q = 27$$

So, $Q_1 = Q_2 = 13.5$ units

And each firm earns a zero economic profit.

If firms collude then $MC = MR$

$$TR = (30 - Q)Q = 30Q - Q^2$$

$$\therefore MR = 30 - 2Q$$

$$\therefore MC = MR$$

$$\therefore 30 - 2Q = 3$$

$$\Rightarrow 2Q = 27$$

$$\therefore Q = 13.5$$

$$\therefore Q_1 = Q_2 = 6.75$$

$$\text{Thus, } \pi = TR - TC$$

$$= (30 \times 13.5) - (3 \times 13.5)$$

$$= 405 - 40.5$$

$$= \{30(13.5) - (13.5)^2\} - \{3 \times 13.5\}$$

$$= 405 - 182.25 - 40.5$$

$$= 182.25$$

$$\text{So, } \pi_1 = \pi_2 = 91.125$$